PHILOMATH
NETWORKS PVT. LTD

6-Days Hands-On Workshop on

"Artificial Intelligence Certification Training Course"

Get ready to earn your Artificial Intelligence Certification with us!

Redefining Learning
Beyond Boundaries

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About Philomath Networks

At Philomath Networks Pvt. Ltd., we specialize in organizing transformative events for schools, organizations, and communities, fostering a culture of exploration, innovation, and growth. From expert-led workshops to competitive events, we create immersive learning experiences that inspire and empower individuals to go beyond boundaries. Partner with us to bring impactful learning to your community!

Partner With Us

Collaborate with Philomath Networks to bring innovation, engagement, and excellence to your institution or organization. Let's work together to create an ecosystem of lifelong learning and impactful experiences!

What Sets Us Apart?

Comprehensive Learning Approach – We blend academic, non-academic, and skill-based experiences through workshops, competitions, expert-led sessions, and immersive programs tailored to real-world applications.
End-to-End Event Management – From conceptualization to execution, we ensure seamless organization of competitions, tech fairs, hackathons, leadership programs, and customized training sessions that align with institutional goals.
Industry & Academic Bridging – We foster collaborations between institutions, corporates, and experts, ensuring that participants gain practical exposure, networking opportunities, and insights into emerging trends beyond traditional education.

Customizable & Scalable Programs – Whether it's a school-level science fest, university hackathon, corporate leadership training, or specialized certification courses, we tailor our events to meet the unique needs of every institution.

Proven Impact & Engagement – Our interactive approach ensures higher participation, deeper learning retention, and enhanced skill development, making every event a transformative experience.

Course
Overview

Course Objective

This intensive training program provides a comprehensive understanding of Artificial Intelligence concepts, tools, and techniques. Participants will explore machine learning, deep learning, natural language processing, and computer vision using Python, TensorFlow, and cloud-based solutions. Through hands-on projects and real-world case studies, learners will gain practical experience in AI model development, optimization, and deployment. By the end of the course, participants will be proficient in building and deploying AI solutions, preparing them for roles in AI, machine learning, and data science.

Course Outcome

- Master AI Basics – Learn core AI, ML, and deep learning concepts.
- Deploy AI Solutions – Optimize and deploy AI models
- Build AI Models – Develop and train ML and deep learning
- Ensure AI Ethics – Address bias, fairness
- Apply NLP & Vision – Use AI for text, speech
- Work on Projects – Build real-world AI applications.

Training Modules

The training modules provide a hands-on learning experience in core AI concepts, machine learning, and deep learning techniques. Participants will begin with AI fundamentals, data preprocessing, and model training using Python and TensorFlow. They will then explore NLP, computer vision, reinforcement learning, and AI ethics to build intelligent systems. Each module is structured around real-world applications, ensuring practical skill development. Additionally, learners will gain expertise in optimizing AI models, managing large datasets, and deploying AI-driven solutions for enhanced automation and decision-making.

Module 1: AI Foundations

- Introduction to AI – History, Evolution, and Applications
- Mathematical Foundations – Linear Algebra, Probability, and Optimization
- Programming for AI – Python, NumPy, Pandas, and AI Libraries
- Data Preprocessing – Feature Engineering, Normalization, and Encoding
- AI Applications – Robotics, Automation, and Predictive Analytics
- AI Tools & Frameworks – TensorFlow, PyTorch, and Scikit-Learn

Module 2: Machine Learning Basics

- Supervised Learning – Regression & Classification Models
- Unsupervised Learning – Clustering & Dimensionality Reduction
- Reinforcement Learning – Basics, Q-Learning, and Applications
- Model Evaluation – Metrics, Cross-Validation, and Bias-Variance Tradeoff
- Feature Selection – Principal Component Analysis and Feature Importance
- Hyperparameter Tuning – Grid Search, Random Search, and Bayesian Optimization

Module 3: Deep Learning & Neural Networks

- Introduction to Neural Networks – Perceptrons, MLP, and Activation Functions
- Convolutional Neural Networks – Image Processing and Object Detection
- Recurrent Neural Networks – Time-Series Analysis and Text Generation
- GANs and Autoencoders – Synthetic Data and Anomaly Detection
- Transfer Learning – Pre-trained Models and Fine-Tuning
- Deep Learning Optimization – Dropout, Batch Normalization, and Regularization

Training
Modules

As the course progresses, participants will explore AI pipeline automation, model deployment, and cloud-based AI solutions. They will learn monitoring techniques using MLflow and TensorBoard and understand security best practices in AI workflows. The final modules focus on AI governance, model optimization, and ethical AI, ensuring learners are industry-ready to implement scalable, secure, and efficient AI-driven solutions. By the end of the training, participants will have the confidence to design, deploy, and manage AI models in real-world environments, preparing them for AI roles in leading tech industries.

Module 4: Natural Language Processing (NLP)

- Text Processing – Tokenization, Stemming, and Lemmatization
- Word Embeddings – Word2Vec, GloVe, and FastText
- Sentiment Analysis – Text Classification and Opinion Mining
- Transformers – BERT, GPT, and Large Language Models
- Speech Recognition – Text-to-Speech and Speech-to-Text Applications
- Chatbots and Conversational AI – Building Virtual Assistants

Module 5: AI Deployment & Ethics

- AI Model Deployment – Cloud AI, Edge AI, and AI Pipelines
- Model Optimization – Quantization, Pruning, and Compression
- AI in Production – MLOps, Monitoring, and Scaling Models
- AI Ethics – Bias, Fairness, and Transparency in AI
- Privacy and Security – Data Protection and Adversarial Attacks
- Regulations and Compliance – GDPR, AI Governance, and Industry Standards

Module 6: AI Capstone Project

- Defining the AI Project – Problem Statement and Dataset Selection
- Data Preparation – Preprocessing, Cleaning, and Augmentation
- Model Development – Training, Testing, and Optimization
- Model Evaluation – Performance Metrics and Improvements
- AI Application Deployment – APIs, Web Apps, and Cloud Hosting
- Certification Assessment – Final Project Submission and Evaluation

Training Schedule & Format

The training schedule is structured as a 6-day intensive AI workshop, providing a hands-on learning experience. Each day focuses on a key AI module, starting with foundational concepts and progressing to advanced topics like machine learning, deep learning, NLP, and computer vision. The workshop includes interactive sessions, hands-on labs, and real-world problem-solving to reinforce learning. Participants will work on mini-projects and practical assignments daily, ensuring a strong grasp of AI tools and techniques. The final day is dedicated to a capstone project, model optimization, and a certification assessment, equipping learners with industry-ready AI skills.

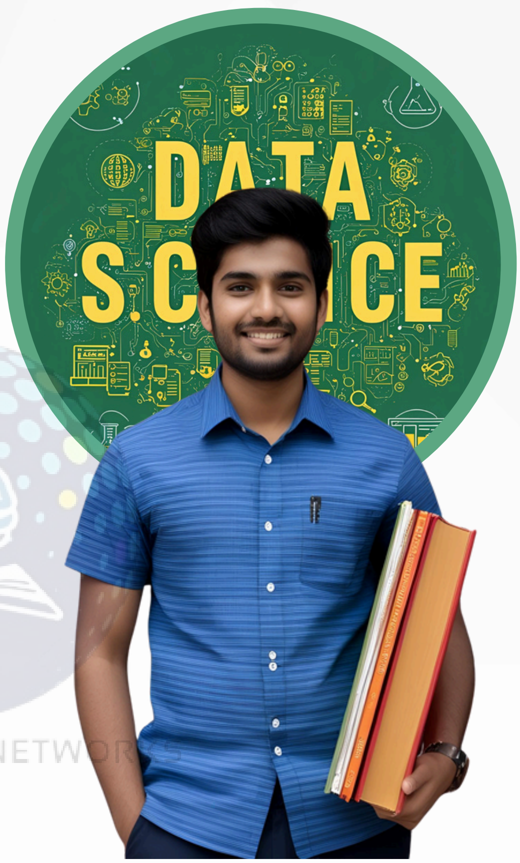
Day	Module	Focus Area	Duration
Day 1	Foundations of Data Science	Data Handling	5hrs
Day 2	Machine Learning Fundamentals	Model Building	5hrs
Day 3	Advanced ML & Deep Learning	Neural Networks	5hrs
Day 4	NLP & Time Series Analysis	Text & Forecasting	5hrs
Day 5	Data Engineering & Big Data Analytics	ETL & Security	5hrs
Day 6	Capstone Project & Certification	Project & Exam	5hrs

Through hands-on exercises and case studies, participants will develop essential AI skills, explore best practices, and tackle real-world AI challenges. The workshop includes live demonstrations, guided projects, and interactive labs to enhance learning. A collaborative format encourages discussions, troubleshooting, and peer learning. By the end, attendees will have a strong AI foundation, enabling them to build, optimize, and deploy scalable AI solutions effectively.

How to Get
Involved

Steps to Enroll in Training Programs

- Contact us via email or phone.
- Share your preferred course, schedule, and any specific requirements.
- Our team will review your request and confirm the booking.
- You will receive confirmation details via email or call .



Contact
Information

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